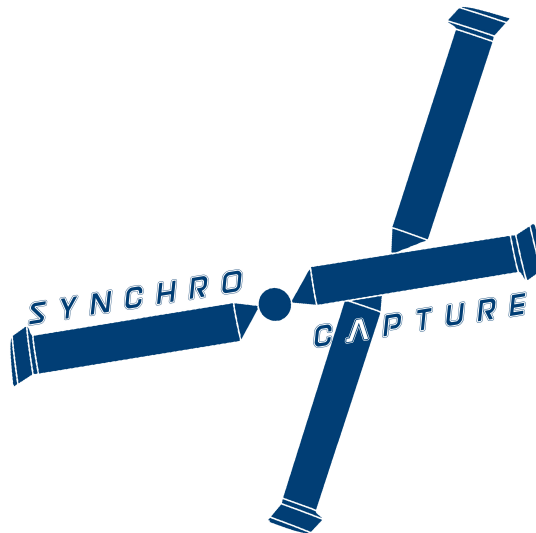


AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

**MODELLING SYNCHROPTER
FLIGHT DYNAMICS
in Aerial Recovery of Rocket Stage**



Student: Filip Bastien
CID: 01336146

Project Supervisor: Dr. Errikos LEVIS
Dr. Andrew WYNN

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Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ

Abstract

A good flight dynamics model is the base for a robust helicopter system that performs well in all flight mode, and allows the control system to respond to any unpredictable disturbances. This report presents the creation of a linear state space model for a novel synchropter concept designed to perform aerial recovery of falling rocket stages up to a mass of 1400kg. The catch manoeuvre consists of matching steep 15° glideslope of the rocket stage suspended under a parafoil and hooking onto it.

The model takes into account the force and moment contributions from the rotors, fuselage, tailplane and dynamics of a slung load. The rotor loads are derived from first principles by developing and running a MATLAB[®] code implementation of Blade Element Momentum Theory. The dynamic flapping motion of the blades is modeled. The loads are linearise around different equilibrium states for hover, cruise and steep descent during the rocket stage catch manoeuvre. Finally this report touches on controller design and assesses the stability and performance of the helicopter model.

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WRR Team Members

Table 1: *Team Members and Roles.*

Name	Team Name	Team Code
Alessandro Bullitta Yassine Ghenima Cyril Gliner	Project co-ordination	WRR01
Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
Anton Le Tejasva Malhotra Atri Sharma Alastair Wood	Structures & CAD	WRR03
Omar Arangath Thomas Cross Zuzanna Janusz Sam Jenner Jo Tan Tom Walker	Systems & Propulsion	WRR04
Filip Bastien Ananya Dubey Kai San Hon Andrew Ng India Tavares	Flight Mechanics, Performance & Control	WRR05
Tom Alston Hakan Serpen Tze Leung Wong	Flight Simulation & Testing	WRR06

†Sub-team leads are shown in **bold**

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1 Introduction

This report is part of a larger project that has the aim of designing an aerial vehicle with the capabilities of aerielly recovering a rocket an orbital rocket stage. It will be briefly discussed why a helicopter was the the most suitable vehicle and what specific helicopter design choices should be made to be best suited for such a mission.

The main topic of the report is modelling the flight dynamics of a synchropter concept designed by other teams within the project. Therefore all necessary parameters to create the model were found in the conceptual design stage and are now known.

The model created is linear, however flight mechanics of helicopter and even more so synchropters are very complex and non-linear, so many assumptions will have to be and the linearisation of the equations of motion might prove particularly challenging.

2 Vehicle Selection - Helicopter

Early on in the project, before many parameters of the study were know, a choice of vehicle had to be made. Different vehicles were considered for the mission, such as a plane, a vtol or a helicopter. This section will make the case for the feasibility, design characteristics and ultimately the use of a helicopter in a mission such as aerielly recovering an orbital rocket stage

2.1 Literature Review

Looking at existing helicopters on the market, a few stood out for their features relevant to the mission. The main points can be taken away:

- Small weight savings dramatically increase service ceiling: a stripped down version of the Eurocopter EC130 landed on the summit of Mt Everest at an altitude of 8,848m, more than doubling its rated service ceiling.
- A skeletal structure allows high payload to empty-weight: The Sykorsky S-64 Skycrane showed the first example of a helicopter with skeletal design having a payload to empty-weight ratio higher than 1.
- Capability of controlling heavy and swinging slung load with large forces acting on it: The Kaman K-Max is shown to wield a 1500kg wrecking ball in a demolition

2.2 Initial Sizing

A first weight sizing for helicopter was done with select payload capacity and hover ceiling, using regression, turboshaft power output scaling and rotor thrust scaling with altitude. This also gave a first Powerplant Sizing and thanks to fuel consumption numbers gave a figure for fuel necessary for determined mission length. See Appendix A for more details.

From this initial sizing the helicopter concept seemed feasible and like the best choice to go forward with.

3 Conceptual Design

Table 2 is a summary main and most useful parameters to come out of the conceptual and preliminary design for an unmanned synchropter capable of aerielly recovering the first and second stage of a rocket. This is the data that the model will be base on. The helicopter must be able to catch the stage and then safely deliver down to the ground[1].

3.1 Mission Profile

1. Takeoff, climb and cruise (Cruise = 185km/h) to recovery altitude and site (20000ft=6096m)
2. Hover at 20000ft
3. Recovery manoeuvre at 35m/s descent at 15°constant glideslope from 20000ft to 3000ft
4. Cruise back (Cruise = 100km/h) with payload to takeoff site
5. Hover and deposit payload, Land

3.2 Vehicle Design Specifications

Table 2: *Summary of Synchropter Concept Specifications*

Parameter	Value
Mass	
Design MTOW	2907 kg
Empty Weight	1229 kg
Fuel Capacity	278 kg
Payload Mass	1400kg
Engine	
Max Power (ASL)	1193kW
Hover Ceiling	6096m
Rotor (for one 2-bladed rotor)	
Rotor Speed Ω	487.5rad/s
Radius R	4.02m
Aspect Ratio.	14
Hinge Offset e	0.15R
Forward Tilt	3°
Side Tilt	14°
Blade Geometry	
Root — Tip	
Coordinates	[e ; 0.9R] — [0.9R ; R]
Chord c	0.287m — 0.287m
Twist θ_{tw}	[-1.8°; -11°] — [-11°; -17°]
Airfoil	NACA23012 — NACA63206
Structure Mass m	3.87kg/m — 4.86kg/m
Velocity	
Cruise Speed	185 km/h, 51 m/s
Cruise Speed (Loaded)	139 km/h, 38 m/s
Max Speed (Tip Mach = 0.85)	225 km/h, 62 m/s
Capture Velocity	126 km/h, 35 m/s

4 State Space Model

A synchropter operates in 6DoF where the motion states are

$$\mathbf{x} = [u, v, w, p, q, r, \phi, \theta, \psi]^T$$

u, v, w are transnational and p, q, r are angular velocities in body fixed axes. ϕ, θ, ψ are Euler angles between the body coordinate system and the inertial frame of reference.

The non-linear equations of motion of the helicopter (Appendix B) are written in terms of these states and the total forces (X,Y,Z) and moments (L, M, N) acting on the helicopter's CG in body axes[2]. These loads are the sum of contributions from both rotors (R_1, R_2), the fuselage (F), the empenage (HT, VT_1, VT_2) and the slung load (S) as seen in figure 1. The following sections will discuss the derivation of kinematics and load contributions from each component and how to linerise them around certain equilibrium points in order to create the state space model for the designed synchropter concept.

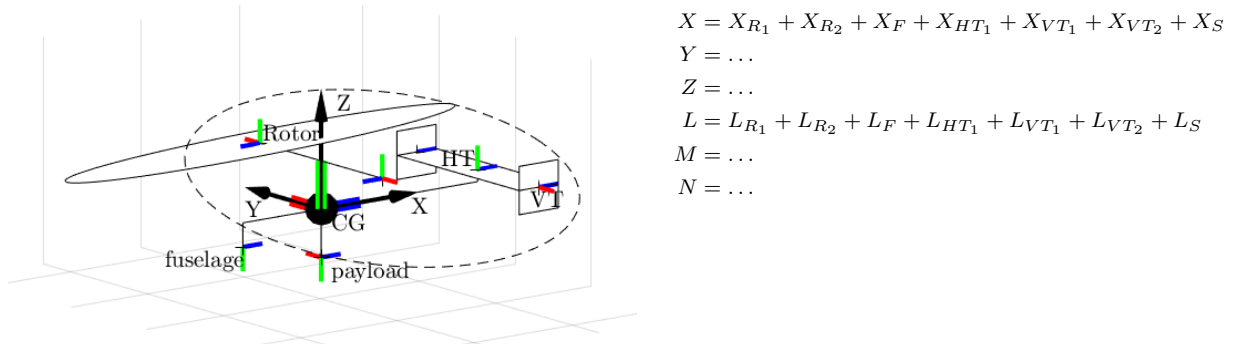


Figure 1: Force and moment contribution

Additional states and equations of motion will be added in order to model the dynamic behaviour of the slung load (Section 4.8) and the rotor blades. The main dynamic rotor behaviour that will be considered is the up-and-down elastic deflection of blades called flapping that is described in more detail in Section 4.5.

4.1 Assumptions

The following list is a summary of the main assumptions that are used in modelling the synchropter's flight dynamics.

4.1.1 Inertial and Structural

- Blade is fully rigid and only deflects due to flapping hinge. Blade lead-lag and torsion are not modelled
- Blade is rigid and no aerodynamic loads are generated inboard of the flap and lag hinges
- Blades rotate at a constant angular speed, Ω . Engine and engine control system dynamics are neglected.
- Swashplate collective and cyclic controls are attached rigidly to the pilot controls.

4.1.2 Aerodynamic

- Calculating induced velocity assumes downstream pressure in the slipstream has equalised to be the same as the initial inlet area pressure.
- The effects of dynamic stall are not included. Past the stall angle, lift goes to zero and drag keeps its maximum value.
- Tip vortex effects are neglected.

4.2 Rotor

The synchropter concept has two side by side counter-rotating and intermeshing 2-bladed rotors. The rotors' rotations are shifted by 90° and they are tilted away from each other so they never collide.

Isolating each rotor and applying a change in coordinate system to non-rotating hub wind axes will simplify the derivation of the rotor loads and kinematics. The hub-wind coordinate system (subscript $\bullet_{tw}$) is aligned with the resultant velocity in the plane of the rotor disc, centered at the hub and has its vertical axes pointing straight up normal to the rotor disc.

Useful coordinate system transformation matrices (T_ϕ , T_θ , T_ψ) presented in Appendix C.2 are used to transform the states from body axes to hub-wind axes. Coordinate transformation is used again to transform the derived rotor forces and moments back into body axes for each rotor.

The rotor can control all degrees of freedom of the helicopter using a combination of the following control states

$$\mathbf{u} = [\theta_0, \theta_{1c}, \theta_{1s}]^T$$

where θ_0 is the blade collective pitch, θ_{1c} is longitudinal cyclic pitch and θ_{1s} is lateral cyclic pitch for rotor 1 (subscript $\bullet_2$ will refer to rotor 2). These are called cyclic controls because they change the blade pitch in a sinusoidal manner with azimuthal position Ψ of the blade such that the blade pitch

$$\theta = \theta_0 + \theta_{1s} \sin \Psi + \theta_{1c} \cos \Psi + \theta_{tw} \quad (1)$$

where the additional θ_{tw} term refers to the built in twist of the blade.

Having transformed the motion states into rotor hub-wind axes and defined the control inputs for the rotor, the rotor kinematics and loads can be derived. This is done by writing and running a Blade Element Momentum Theory code for the designed rotor.

4.3 Blade Element Momentum Theory Code (BEMT)

A Blade Element Momentum Theory code derives rotor loads and kinematics from first principles by first analysing the aerodynamic and inertial loads on any generic blade section at any radial or azimuthal position. Momentum theory is used to accurately determine the velocity and angle of the flow past this section giving the aerodynamic forces. Inertial forces are due to the rotation and other motions of the blade section.

Figure 2 shows how the velocities and angles will be defined on a generic blade section.

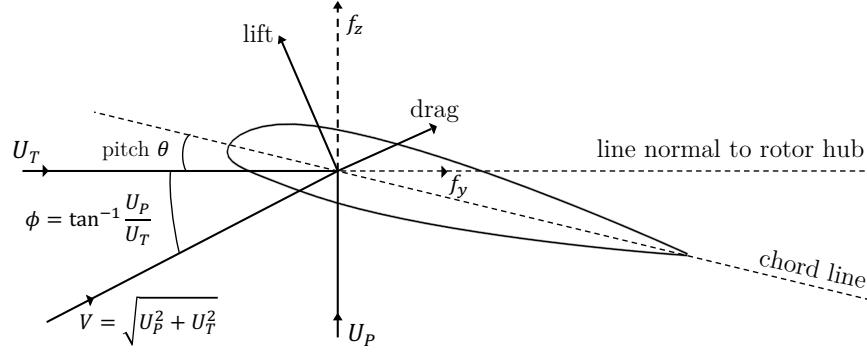


Figure 2: Blade section angles and local velocities

For a generic blade section of radial position r_b and azimuthal position Ψ :

The oncoming velocity V has in-plane component U_T and normal component U_P . U_T is equal to the velocity produced by rotating motion with additional increments due to the oncoming wind, and angular velocity transformed to the rotating system (ω defined in Appendix [ROTATING TRANSFORMATION])

$$U_T = r_b \Omega + u_{hw} \sin \Psi + \beta \frac{r_b}{R} \omega_x \quad (2)$$

U_P is equal to the rotor induced velocity v_i with additional increments due to climb speed w_{hw} , flapping velocity $\dot{\beta}$ and rotor/vehicle forward speed u_{hw} . Calculating v_i accurately it is essential in determining the angle of attack which in turn gives the aerodynamic forces on the blade and is discussed in Section [Inflow Models].

$$U_P = w_{hw} - v_i - \beta u_{hw} \cos \Psi + \frac{r_b}{R} \omega_y - r_b \dot{\beta} \quad (3)$$

ϕ is the angle of the resulting oncoming velocity to the rotor plane

$$\phi = \arctan \frac{U_P}{U_T} \quad (4)$$

With the velocity and angle of attack $\theta + \phi$, the aerodynamic forces can be determined

$$\text{lift} = \frac{1}{2} \rho (U_T^2 + U_P^2) c C_{L(\theta+\phi)} \quad (5)$$

$$\text{drag} = \frac{1}{2} \rho (U_T^2 + U_P^2) c C_{D(\theta+\phi)} \quad (6)$$

where the lift and drag coefficients are a function of angle of attack presented in table 3. If the angle of attack exceeds the defined range of the functions, the blade has stalled and thus the lift is set to zero and drag takes its maximal value. The aerodynamic resultants are the, normal thrust f_z and the in-plane torque f_y

$$f_z = -\text{lift} \cos \phi - \text{drag} \sin \phi \quad (7)$$

$$f_y = \text{drag} \cos \phi - \text{lift} \sin \phi \quad (8)$$

Now taking into account the inertial loads, the approximate acceleration components at the blade section are given by the expressions

$$a_{xb} = r_b (-\Omega^2 + 2\dot{\beta} \omega_y - 2\Omega \beta \omega_x) \quad (9)$$

$$a_{yb} = r_b(-\beta(q_{hw} \sin \Psi - p_{hw} \cos \Psi) + r_{hw}\beta\omega_y) \quad (10)$$

$$a_{zb} = r_b(2\Omega\omega_x + (q_{hw} \cos \Psi + p_{hw} \sin \Psi) - r_{hw}\omega_x - \Omega^2\beta) \quad (11)$$

Finally, the derived expressions sectional loads are written as [3]

$$dX_R = -(f_z - ma_{zb})\beta \cos \Psi - (f_y - ma_{yb}) \sin \Psi + ma_{xb} \cos \Psi \quad (12)$$

$$dY_R = -(f_z - ma_{zb})\beta \sin \Psi - (f_y - ma_{yb}) \cos \Psi + ma_{xb} \sin \Psi \quad (13)$$

$$dZ_R = f_z - ma_{zb} + ma_{xb}\beta \quad (14)$$

$$dL_R = r_b(dZ \sin \Psi + dY\beta) \quad (15)$$

$$dM_R = r_b(dZ \cos \Psi + dX\beta) \quad (16)$$

$$dN_R = r_b(dX \sin \Psi + dY \cos \Psi) \quad (17)$$

In order to derive the total resulting forces and moments of the rotor at the hub, the blades are discretised into radial elements (dr_b) and the rotation is discretised into azimuthal elements ($d\Psi$). The sectional loads are calculated then integrated numerically along the length of the designed blade, for each blade and averaged over a rotor revolution. The full code implementation of the BEMT method with an iterative technique for determining the induced velocity can be found in Appendix D.

Figure 3 shows the results of the BEMT code giving blade thrust for each section and angle of attack with states set to a cruise equilibrium.

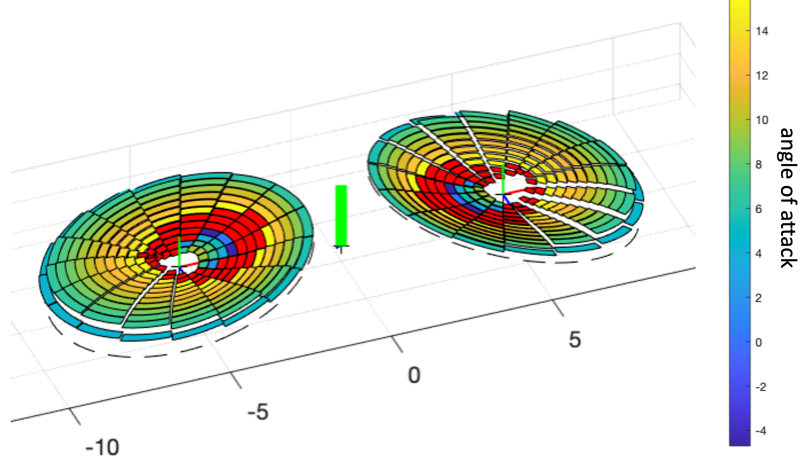


Figure 3: BEMT code results with deflections analogous to element vertical force and color representing blade angle of attack in forward flight (red: stall)

4.4 Inflow Models

Calculating the rotor inflow, or induced velocity v_i is the most challenging part of the rotor aerodynamics to model accurately. Rotor inflow depends on the rotor's thrust and velocity distribution which are not uniform over the rotor disc. Many other complicated effects such as trailing vortex interactions modelled in the Free-Vortex Wake model[REF Free Vortex Wake] are ignored in this section.

Table 3: *Blade Section Aerodynamic Coefficients*

Airfoil	NACA23012	NACA63206
$C_L(\alpha_{\text{rad}}) = p_1\alpha^3 + p_2\alpha^2 + p_3\alpha + p_4$		
p_1	-29.1271	-92.0013
p_2	0.4356	-2.4531
p_3	7.0548	6.8650
p_4	0.1392	0.1711
$C_D(\alpha_{\text{rad}}) = p_1\alpha^4 + p_2\alpha^3 + p_3\alpha^2 + p_4\alpha + p_5$		
p_1	9.1910	82.8051
p_2	-0.2915	-3.2461
p_3	-0.1336	1.7297
p_4	-0.0102	-0.0184
p_5	0.0090	0.0029
$\alpha_{\min}$	-13.00°	-7.75°
$\alpha_{\max}$	+15.75°	+7.50°

4.4.1 Uniform Inflow

The first inflow model is a uniform distribution of induced velocity over the disc based on the total thrust produced by the rotor. The uniform model is based on simple momentum theory over the whole area of the rotor disc.

$$v_0 = \sqrt{\frac{T}{2\pi\rho R^2}} \quad (18)$$

It easy to determine a value of the induced velocity in hover for example as the total thrust produced should be equal to the weight of the helicopter. This uniform inflow is used as an initial guess for the iterative method described below.

4.4.2 Linear Models

Different linear inflow models from Leichman's *Definition of the Approximate Induced Velocity Field* ([4] Section3.5.2) were considered. Each model corresponds to a different flight mode (hover, axial flight, forward flight) and is linear along the x and y axes of the rotor disc. The induced velocity distribution is written in the general form

$$v_i = v_0(1 + k_x \frac{r_b}{R} \cos(\Psi) + k_y \frac{r_b}{R} \sin(\Psi)) \quad (19)$$

where k_x and k_y are weighting factors that represent the deviation of the inflow from the uniform value predicted by simple momentum theory in equation [LABEL EQUATION UNIFORM].

4.4.3 Iterative Method from Momentum Theory

To calculate v_i more accurately, an axial flow momentum balance must be applied to predict the induced effects on every blade element (dr_b). A major complexity in applying this theory comes from the fact that the induced velocity is a function of the thrust produced by each element which is in turn based on the effects produced by the induced velocity itself, hence the equations for v_i become non-linear and why it must be solved with an iterative method. With cyclic variations

of thrust in certain flight modes, the process is also run on azimuthal elements of the disk ring ($d\Psi$).

The iteration is done for every radial and azimuthal element increment in the BEMT code to find an accurate value of the induced velocity and has the following structure:

1. Guess an initial value of $v_i = v_0$.
2. Find the flow angle of attack on the blade
3. Estimate element thrust f_z
4. Momentum change equilibrium is used to improved the estimate of v_i

$$v_{i|new} = 0.9v_i + 0.1 \frac{4f_z}{\pi \rho r_b U_P} \quad (20)$$

5. Process is repeated until value of v_i has converged

The convergence for this nonlinear system of equations is not guaranteed, however, a simplified Crank-Nicholson convergence enhancing technique was employed. A full implementation of the method can be found in the BEMT Code (Appendix [BEMT CODE]) and the momentum theory is taken from the derivation for rotor thrust and torque in [5].

4.5 Flapping

The high aspect ratio of the synchropter blades along with the high loading they experience means inherently deform elastically. On the top of this a physical flapping hinge allows the blade to move in an up and down motion to alleviate some of the load transfer to the fuselage. In our model the blade is assumed rigid and can only deflect due to the moments at the modelled flapping hinge located at $r_b = e = 0.15R$. In order to stay within design parameters and avoid hitting the opposite hub the blade must not flap more than 12° . From structural analysis the maximum bending moment at the flapping hinge is $25.2 \cdot 10^5 \text{Nm}$ sizing the model's flapping hinge spring stiffness to $K_\beta = 12 \text{MN/rad}$.

Flapping dynamics are important as they change the direction of the thrust vector by flapping one blade up and the opposite blade down, in essence tilting the rotor. This improves handling capabilities and has a stabilising effect reducing the dissymmetry of lift in forward flight and damping roll response. For example, in response to a roll rate to the right, blade flap tilts the rotor to the left to provide the desired roll damping of the aircraft.

For any azimuthal position Ψ the flapping equation was derived by summing the moments at the flapping hinge [6]

$$K_\beta \beta = \int_e^R dZ(r_b - e) + \beta r_b (dX \cos \Psi - dY \sin \Psi) dr_b \quad (21)$$

Flapping angle is assumed to vary in a sinusoidal manner with azimuthal position Ψ such that

$$\beta = \beta_{1s} \sin \Psi + \beta_{1c} \cos \Psi \quad (22)$$

Adding flapping dynamic to the model requires adding flapping angles β_{1c} and β_{1s} as states and adding two equations of motion by evaluating the flapping equation (equation 21) at $\Psi = 0^\circ$ and $\Psi = 90^\circ$ for each rotor.

4.6 Fuselage

The fuselage shape was designed to minimise aerodynamic drag and downforce in order to keep forward flight as efficient as possible. The synchropter concept has pitch dependant aerodynamic coefficients (in units m^2) determined using CFD by Alessandro in his aerodynamic analysis of the fuselage [7]. The pitch angle θ is the angle between the wind and body axes.

$$X_F = \frac{1}{2}\rho u^2 C_D \quad (23)$$

$$Z_F = \frac{1}{2}\rho u^2 C_L \quad (24)$$

$$M_F = \frac{1}{2}\rho u^2 C_M \quad (25)$$

Table 4: *Fuselage aerodynamic coefficients, can be linearly interpolated between given pitch angle θ*

θ	C_D	C_L	C_M
0°	0.1645	-0.0714	0.1541
6°	0.1775	-0.0965	0.1196
12°	0.1962	-0.1483	0.0605

4.7 Empenage

The empenage of the synchropter concept is an all-moving h-tail design with additional control input δ_e for the elevator and a common control input δ_R for both rudders. The forces for the empenage interact with the rotor wake and are derived by Kevin in his aerodynamic design and analysis of the empenage [8].

4.8 Slung Load

The payload is modelled as a spherical pendulum with two degrees of freedom (θ_{PL}, ϕ_{PL}) , enabling payload oscillation analysis in the lateral and longitudinal planes. Cable mass and drag are neglected but payload drag is calculated and incorporated. Induced forces and moments on the synchropter due to payload oscillations are then added to the helicopter full state model, resulting in 4 additional states identifying payload motion.

$$[\theta_L, \phi_L, \dot{\theta}_L, \dot{\phi}_L]^T$$

The full derivation of these forces and equations modelling payload dynamics and payload/vehicle interactions are laid out in Ananya Dubey's Report [9].

After a dynamic stability analysis of the system it was shown that in general the coupling between modes is more significant during hover with lateral oscillations being quite dominant. During cruise phase, longitudinal and lateral payload oscillations interact less with helicopter modes.

4.9 Equilibrium and Trim

Expressions derived for load contributions from both tilted rotor, the fuselage, empenage and slung payload are added together and the resulting loads at the CG were found. The force resultant (X,Y,Z) was found by adding all forces in body axes. The moment resultant (L,M,N) was found

by adding all moments as well as all forces in body axes multiplied by their respective (dx,dy,dz) moment arms to the CG.

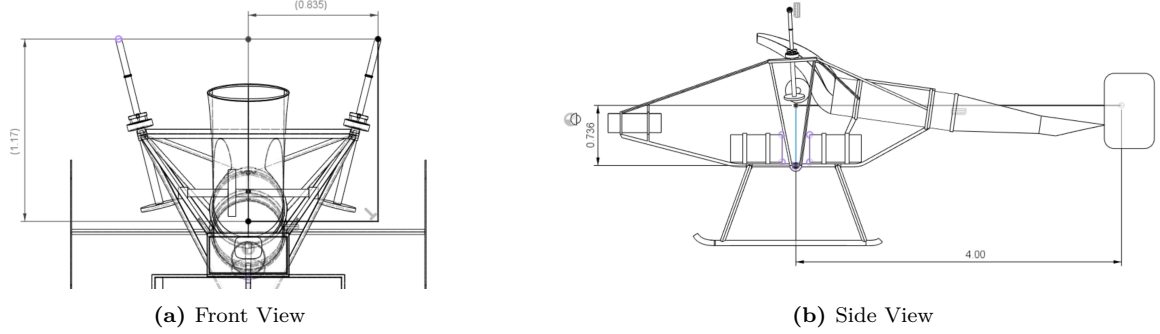


Figure 4: Synchropter geometry and moment arms

The most constraining steady flight modes necessary for the mission are the following. these become more constraining in terms of performance as altitude increases and air density decreases:

- Hover (loaded/unloaded) - 0m/s
- Cruise (loaded/unloaded) - 51m/s (unloaded)
- Descend at constant 15°glide slope (loaded/unloaded) - 35m/s

For any of these steady stages of flight, physical equilibrium can be found for states $[u, v, w, p, q, r, \phi, \theta, \psi]^T$ for all stages of flight. Equilibrium values for flap angles $[b_{1c}, b_{1s}, b_{2c}, b_{2s}]^T$ are found solving the flapping equation (equation 21) in the lateral and longitudinal directions respectively for both blades. Equilibrium can also be determined for payload angles [9] when applicable.

Finally, an iterative solver to find the control inputs for trimmed flight in each studied flight mode was developed, making use of the Levenberg–Marquardt optimisation algorithm. A full implementation of this solver is described in Andrew Ng’s report[10].

4.10 Linearisation

The expressions for the loads are linearised around the equilibrium states (or trim control inputs) for each flight mode using a first order Taylor expansion. Assuming the linearisation of the loads is valid (steady flight modes) they can be written in the general form

$$X = X_e + \frac{\partial X}{\partial u} \delta u + \frac{\partial X}{\partial v} \delta v + \dots + \frac{\partial X}{\partial \theta_0} \delta \theta_0 + \frac{\partial X}{\partial \theta_{1c}} \delta \theta_{1c} + \dots \quad (26)$$

where $\frac{\partial X}{\partial \mathbf{x}}$ are stability derivatives and $\frac{\partial X}{\partial \mathbf{u}}$ are stability derivatives. These derivatives are found using central difference scheme with a small perturbation in the state (or control inputs)

$$\mathbf{x} = \mathbf{x}_e + \delta \mathbf{x} \quad (27)$$

where the a given state states $\mathbf{x}$ are is perturbed from its equilibrium value $\mathbf{x}_e$ by a perturbation of size $\delta \mathbf{x}$.

The derivatives were evaluated with with a perturbation of $\pm 10^{-1}$ in all states and control inputs.

The linearised force and moment expressions with their stability and control derivatives are plugged back into the synchropter equations of motion (Appendix B) evaluated at equilibrium states.

This linearised set of equations is written in State Space form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

where the state matrix $\mathbf{A}$ and control matrix $\mathbf{B}$ are written in terms of 17 states:

$$\mathbf{x} = [u, v, w, p, q, r, \phi, \theta, \psi, b_{1c}, b_{1s}, b_{2c}, b_{2s}, \theta_L, \phi_L, \dot{\theta}_L, \dot{\phi}_L]^T$$

and 7 control states:

$$\mathbf{u} = [\theta_0, \theta_{1c}, \theta_{1s}, \theta_{2c}, \theta_{2s}, \delta_e, \delta_R]^T$$

4.11 Derivative analysis

The X, Y, Z forces and L, M, N moments can also be evaluated for a range of perturbations sizes in all states and control inputs in order to see how their derivatives vary with perturbation size.

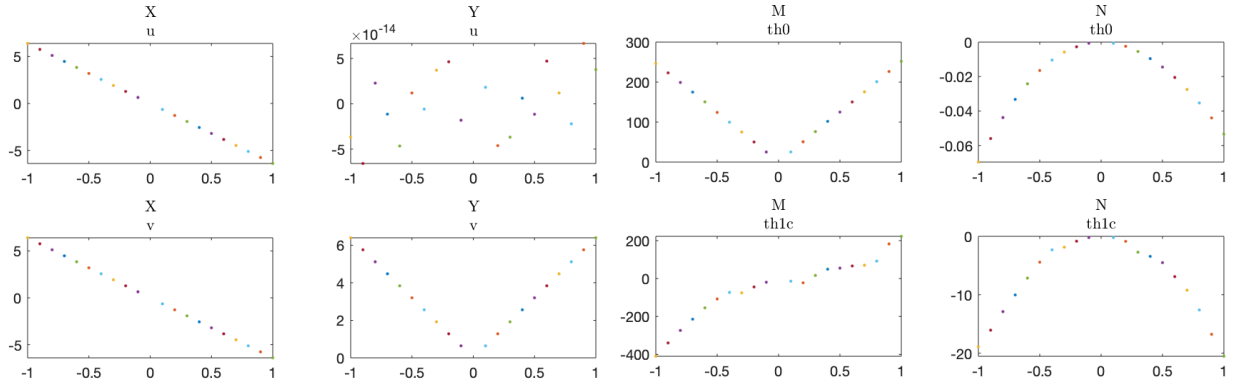


Figure 5: Load vs. perturbation size plots shows the change of stability and control derivatives with change in perturbation size

5 Controller Design

The control methodology chosen was LQQ control which consists of a Kalman filter and a LQR. The Kalman filter would allow to give more accurate estimation of the states combining data from onboard measurements with the predictions from the model giving outputs with less noise and uncertainty. The LQR part of the controller would be used to place the eigenvalues of the closed loop system in order to achieve the desired properties. A more detailed look into this method and design of the controller is detailed in Ananya's [9] and India's [11] reports.

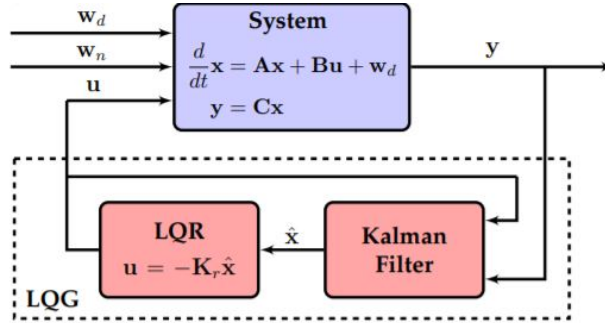


Figure 6: LQG controller block diagram

This type of controller requires the system to be observable/controllable, or at the very least detectable/stabilisable, however this was not the case, hence the performance of the controller could not be assessed. This lack of observability and controllability could be due to issues in the derivation of the stability derivatives, and hence the linearisation of the system. The presence of highly non-linear dynamics and interaction with the payload might make linearisation of the model invalid/impossible and the need for the use of non linear control methods such as Model Predictive Control.

6 Stability and Performance

The full stability and performance analysis were done in Andrew's and Jason's reports [10][12]. However the main takaways are:

- At hover, the helicopter is statically stable with respect to speed, pitch, roll and yaw
- At cruise, it is statically stable with respect to speed and pitch.
- Dynamic stability analysis reveals an unstable phugoid mode and dutch roll mode at hover, characteristic to most rotorcraft
- the unstable phugoid mode becomes stable at our design cruise speed of 30ms-1 and an altitude of 3000ft
- The performance of the synchrocopter is analysed using the data provided from the advanced rotor BEMT code developed in the creation of the synchrocopter model
- The results show that our helicopter is capable of achieving our mission profile with our current power sizing and rotor design

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Appendices

A Initial Weight Sizing from Physics and Regression Data

B Synchropter Equations of Motion

B.1 Basic Helicopter Equations of Motion

$$\dot{u} = -(wq - vr) + \frac{X}{M_a} - g \sin \theta \quad (28)$$

$$\dot{v} = -(ur - wp) + \frac{Y}{M_a} + g \cos \theta \sin \phi \quad (29)$$

$$\dot{w} = -(vp - uq) + \frac{Z}{M_a} + g \cos \theta \cos \phi \quad (30)$$

$$\dot{p} = I_{xx}^{-1}(L + qr(I_{yy} - I_{zz})) \quad (31)$$

$$\dot{q} = I_{yy}^{-1}(M + pr(I_{zz} - I_{xx})) \quad (32)$$

$$\dot{r} = I_{zz}^{-1}(N + pq(I_{xx} - I_{yy})) \quad (33)$$

$$\dot{\phi} = p + q \sin \phi \tan \theta + r \cos \phi \tan \theta \quad (34)$$

$$\dot{\theta} = q \cos \phi - r \sin \phi \quad (35)$$

$$\dot{\psi} = q \sin \phi \sec \theta + r \cos \phi \sec \theta \quad (36)$$

$$(37)$$

B.2 Flapping Equations of Motion for 2 Rotors

$$\dot{\beta}_{1s} = f(u, v, w, p, q, r, \beta_{1s}, \beta_{1c}, \theta_0, \theta_{1s}, \theta_{1c}) \quad (38)$$

$$\dot{\beta}_{1c} = g(u, v, w, p, q, r, \beta_{1s}, \beta_{1c}, \theta_0, \theta_{1s}, \theta_{1c}) \quad (39)$$

$$\dot{\beta}_{2s} = h(u, v, w, p, q, r, \beta_{2s}, \beta_{2c}, \theta_0, \theta_{2s}, \theta_{2c}) \quad (40)$$

$$\dot{\beta}_{2c} = j(u, v, w, p, q, r, \beta_{2s}, \beta_{2c}, \theta_0, \theta_{2s}, \theta_{2c}) \quad (41)$$

B.3 Slung Payload Equations of Motion

$$\dot{\phi}_L = \dot{\phi}_L \quad (42)$$

$$\dot{\theta}_L = \dot{\theta}_L \quad (43)$$

$$\ddot{\phi}_L = k(u, v, w, p, q, r, \phi, \theta, \phi_L, \theta_L) \quad (44)$$

$$\ddot{\theta}_L = m(u, v, w, p, q, r, \phi, \theta, \phi_L, \theta_L) \quad (45)$$

C Coordinate Syetems

C.1 Non-Rotating Hub Wind Axes

C.2 Coordinate System Transformation

Matrices

$$T_\phi \tag{46}$$

$$T_\theta \tag{47}$$

$$T_\psi \tag{48}$$

$$\tag{49}$$

figure

- Transforming body-axes states to non rotating hub-wind axes

$$[u_{hw}, v_{hw}, w_{hw}] \tag{50}$$

- Transforming angular velocities to rotating system
- Transforming hub-wind forces back to body-axes

D BEMT Code Implementation

This script evaluates all of something and plots something

```

1  %-----
2  % AERO96005 GROUP DESIGN PROJECT
3  % Numeric BEMT Code implementation
4  %Filip Bastien
5  %15/06/2020
6  %-----
7  clear
8  clc
9  close all
10
11 %input parameters
12 g=9.81; W=1500*g; alt=4870; [tau,tau,tau,rho]=atmosisa(alt); R=4.02; ...
    Omega=487.5*(2*pi)/60; PI=pi; AR=14; e=0.15*R; c=R/AR; Nb=2; b=0; b_d=0;
13 %blade tilt
14 rotortilt.th=deg2rad(10);
15 rotortilt.phi=deg2rad(0);
16
17 %ste velocities
18 u=35*cosd(15);
19 v=0;
20 w=-35*sind(15);
21
22 %coordinate transformation to hub axes
23 Vh = T_phi(-rotortilt.phi)*T_th(-rotortilt.th)*[u;v;w]; ...
    %[u-q*hr;v+p*hr+r*xcg;w-q*xcg]
24 [uh,vh,wh] = deal(Vh(1),Vh(2),Vh(3));
25
26 %coordinate transformation to hub-wind axes
27 uhw = sqrt(uh^2+vh^2);
28 whw = wh;
29 sideslip = acos(uh/uhw);
30
31 %aerodynamic coefficients
32 Cl1 = @(x) - 29.1271*x^3 + 0.4356*x^2 + 7.0548*x + 0.1392;
33 Cl2 = @(x) - 92.0013*x^3 - 2.4531*x^2 + 6.8650*x + 0.1711;
34 Cd1 = @(x) 9.1910*x^4 - 0.2915*x^3 - 0.1336*x^2 - 0.0102*x + 0.0090;
35 Cd2 = @(x) 82.8051*x^4 - 3.2461*x^3 + 1.7297*x^2 - 0.0184*x + 0.0029;
36 aoamin1=deg2rad(-13); aoamax1=deg2rad(22);
37 aoamin2=deg2rad(-7.75); aoamax2=deg2rad(11);
38
39 %blade twist
40 th.tw1 = @(x) x/(0.9*R)*deg2rad(-11); %linear twist angle
41 th.tw2 = @(x) deg2rad(-17+11)*(x-0.9*R)/(R-0.9*R)+deg2rad(-11); %linear twist angle
42
43 %Thrust
44 T=W;
45
46 %Constant Inflow
47 vih=sqrt(T/(2*PI*rho*R^2));
48
49
50 %% discretisation
51 %azimuth discretisation
52 npsi = 16;

```

```

53 dpsi = 360/npsi;
54 psi_array = transpose(0:dpsi:360-dpsi);
55
56 %radial discretisation
57 nrb = 16;
58 dr = (R-e)/nrb;
59 rb_array = e+dr/2:dr:R-dr/2;
60
61
62 %% Solver
63 thlc=deg2rad(0);
64 thls=deg2rad(0);
65
66 %% Vertical Equilibrium Iteration
67 th0_array=deg2rad(-90):deg2rad(1):deg2rad(90);
68 for k=1:181 %while Zerror>1 %
69     th0=th0_array(k);
70
71     %initialise forces and moments
72     X=0; Y=0; Z=0; L=0; M=0; N=0;
73
74     %Azimuth increment
75     for i = 1:npsi
76         psi = psi_array(i); %azimuth position
77         cpsi=cosd(psi);
78         spsi=sind(psi);
79         %Radial increment
80         for j = 1:nrb
81             rb = rb_array(j); %radial position
82
83             if rb<0.9*R %Root section
84                 Cl = Cl1;
85                 Cd = Cd1;
86                 th_tw = th_tw1(rb);
87                 aoamin=aoamin1;
88                 aoamax=aoamax1;
89             elseif rb>= 0.9*R %Tip section
90                 Cl = Cl2;
91                 Cd = Cd2;
92                 th_tw = th_tw2(rb);
93                 aoamin=aoamin2;
94                 aoamax=aoamax2;
95             end
96
97             Ut=Omega*rb+uhw*cos(b)*spsi;
98             th=th_tw+th0+thlc*cpsi+thls*spsi;
99
100
101         %% Inflow Velocity Iteration
102         %guess initial value of induced velocity
103         vi=vih;
104
105         %iteration logical variable, count and check flag
106         finished=false; sum=1; itercheck=0;
107         while (~finished)
108             %normal velocity components
109             Up=vi+whw+uhw*sin(b)+b_d*rb*spsi;
110             %flow angle
111             phi=atan(Up/Ut);

```



```

112         %blade angle of attack
113         aoa=phi+th;
114
115         %aerodynamic forces
116         lift = 1/2*rho*(Ut^2+Up^2)*c*Cl(aoa)*Nb/npsi;
117         drag = 1/2*rho*(Ut^2+Up^2)*c*Cd(aoa)*Nb/npsi;
118
119         %thrust grading
120         fz=lift*cos(phi)+drag*sin(phi);
121         fy=drag*cos(phi)-lift*sin(phi);
122
123         %momentum check on induced velocity
124         tem1=fz/(pi/4.0*rb*rho*Up);
125         %stabilise iteration
126         vinew=0.9*vi+0.1*tem1;
127         if vinew<0
128             vinew = 0;
129         end
130
131         %check for convergence
132         if (abs(vinew-vi)<1.0e-5)
133             finished=true;
134         end
135
136         %new inflow velocity
137         vi=vinew;
138
139         %increment iteration count
140         sum=sum+1;
141         %check to see if iteration stuck
142         if (sum>500)
143             finished=true;
144             itercheck=1;
145         end
146     end
147
148     %check for stall
149     if aoa<aoamin || aoa>aoamax
150         fz=0;
151         fy=0;
152     end
153
154     %% Integration over radius and rotation
155     x=fy*spsi;
156     y=-fy*cpsi;
157     z=fz;
158
159     X = X+x*dr;
160     Y = Y+y*dr;
161     Z = Z+z*dr;
162     L = L+(-z*spsi*cos(b)+y*sin(b))*rb*dr;
163     M = M+(-z*cpsi*cos(b)+x*sin(b))*rb*dr;
164     N = N+(x*spsi+y*cpsi)*cos(b)*rb*dr;
165
166     val(j,i)=rad2deg(phi);
167     val(j,i)=fz;
168
169 end
170 end

```

```
171     hold on
172     plot(rad2deg(th0),Z,'k.')
173     if Z>W
174         rad2deg(th0)
175         break
176     end
177 end
178 %RESULTS
179 Rrotor_Forces = T_phi(rotortilt_phi)*T_th(rotortilt_th)*T_psi(-sideslip)*[X;Y;Z];
180 Rrotor_Moments = T_phi(rotortilt_phi)*T_th(rotortilt_th)*T_psi(-sideslip)*[L;M;N];
```